Sanskar Jain

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Summary

Backend and distributed systems engineer with 2+ years of experience building cloud-native, event-driven infrastructure at Verisk Analytics. Deep expertise in designing fault-tolerant, high-throughput distributed systems using **Kafka**, **AWS**, and statically typed languages (**Java**, **C#**). Proven track record shipping large-scale, production systems with strong ownership, observability, and end-to-end reliability across cloud environments.

Skills

Languages: Java, C#, Python, Go (familiar)
Distributed Systems: Apache Kafka, Event-Driven Architecture, CQRS, Pub/Sub, Idempotent Processing, Retry/Backoff, DLQ
Cloud & Infrastructure: AWS (EKS, ECS, Lambda, S3, SNS, SQS, ElastiCache), Docker, Kubernetes, Terraform, CI/CD
Networking & Security: Kong API Gateway, OAuth 2.0, JWT, Okta, REST APIs, API Versioning
Databases & Storage: MongoDB, PostgreSQL, MySQL, Redis, Indexing, Query Optimisation
Observability & Tools: OpenTelemetry, Splunk, xUnit/NUnit, Swagger/OpenAPI, Azure DevOps, Jira

Experience

Verisk Analytics Jan 2024 – Present
Software Engineer II, Hyderabad, India

- Co-designed **RulesRunner**, a stateless distributed microservice using **CQRS** and **SQS** that decoupled compliance processing from clients, reducing end-to-end latency from **3000ms to 500ms (83%)** with horizontal scalability.
- Architected **event-driven pipelines** with idempotent message processing, configurable retry/backoff strategies, dead-letter queues (DLQ), and **SNS**-based fan-out for distributed, fault-tolerant consumers.
- Engineered **AWS Lambda** serverless processing pipelines with zero message loss guarantees under network partitions and failure conditions, aligning with production SLA requirements.
- Built large-scale async data transfer backend for **Xactimate** handling **30+ data types** via **S3 presigned URLs** and **MongoDB**, reducing transfer time by **38%** and data integrity issues by **25%**.
- Deployed and operated services on **AWS EKS** and **ECS** with container orchestration using **Kubernetes** and **Docker**, supporting rolling deployments and zero-downtime releases.
- Built **ASP.NET Core** REST APIs handling **200K+ weekly requests**; implemented URL-based versioning for seamless backward-compatible client migrations.
- Improved hot-path API response time by **34%** through **Redis/ElastiCache** caching, connection pooling, and targeted **MongoDB / PostgreSQL** query and index optimisation.
- Authored **production-safe, idempotent migration scripts** enabling zero-downtime schema deployments across distributed database clusters.
- Accelerated deployment cycles by **50%** and build time by **33%** by re-architecting **Azure DevOps** pipelines with staged gates, parallel jobs, and automated integration tests.
- Instrumented services with **OpenTelemetry** distributed tracing and **Splunk** dashboards to achieve deep observability, reducing MTTD for production incidents.

Projects

Kafka-Driven Expense Tracker – Distributed Microservices Platform Aug 2023 – Dec 2023

- Built a cloud-native platform with **5+ Spring Boot (Java) microservices** communicating via **Apache Kafka** topics for real-time, async event streaming between producers and consumers.
- Designed **Kafka** producers/consumers with offset management and consumer group coordination; containerised with **Docker** and orchestrated via **Kubernetes**, simulating **1000+ concurrent transactions** for load testing.

Certifications

AWS Certified Cloud Practitioner Amazon Web Services — Oct 2024

Education

B.Tech in Computer Engineering – Vishwakarma Institute of Technology, Pune 2024 — CGPA: 8.8